

Harassment Detection (Thai Text)

Presented to

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Introduction

In the digital age, where we spend most of our time on social media platforms, communication is faster, easier, and more accessible than ever, and so is the harassment that comes from social media platforms as well[1].

We have seen preventive measures that have worked in screening the harassment messages of the English text, but it isn't the same case for the harassment messages with Thai text. Thai texts have many nuances that make it difficult to capture the author's true intentions behind their message. Some messages that may sound like flat-out harassment might be sarcasm or slang that translates to a whole other meaning.

For this project, we will develop a Harassment detection system, specifically with Thai text. We sourced the dataset from a public github repositories that is called ThaiToxicityTweetCorpus[2], which contains a 44-word dictionary of harassment words, and is sufficient for the scope of our project. The project pipeline involves data loading, text processing and tokenization, vectorization, and then training the model.

From the perspective of the operating system, all the steps of this project will heavily utilize the computational power of our device. The data loading will relate to the I/O Management and the file system, with the potential problem of an I/O bottleneck. The text processing and tokenization will relate to the process management and CPU scheduling, with the problem of CPU starvation and slow performance. The vectorization of the tokenized text will relate to the memory management, with the potential problem of memory shortage. Lastly, the model training will relate to the CPU scheduling, with the potential problem of slow performance.

Objective

Problem Statement

Cyberbullying on Thai social media, such as X, has grown beyond control. Current English-based models struggle with Thai linguistic complexities, such as the lack of word boundaries and evolving slang, leading to a critical shortage of real-time detection systems. This project develops an automated Thai-centric detection pipeline to address this gap while utilizing the high-load processing requirements to evaluate and optimize core Operating System resource management principles.

This project aims to achieve the following objectives:

1. Load the data from ThaiToxicityTweetCorpus into the system, and use data buffering to load many large blocks of text. The datasets from GitHub are already cleaned and processed, so we will focus on avoiding I/O bottlenecks caused by constant hardware interruptions. This process will test the I/O handling and file system performance when retrieving a huge amount of data.

2. The Text Processing and Tokenization by splitting the workload across multiple CPU cores with the multiprocessing library. Because this part will take up huge computational resources, we will split the processing among multiple workers. This process will exercise the process of CPU scheduling by finding the middle point between high throughput and the overhead that is caused by too much context switching.

3. The Vectorization of the data, which converts text strings into numerical matrices by using batch processing or data generators. If we don't load all transformed data into the RAM at the same time can prevent hitting the memory's physical limits. This will ensure that we can stay in the scope of efficient memory management and avoid stimulating the slow OS swapping or aggressive page replacement algorithms.

4. Train the categorized model to identify and flag toxic Thai content. While running the training script as a background task with lower priority. This lets us see how the OS manages CPU usage by prioritizing user tasks over heavy background computation, ensuring that the computer remains smooth and responsive.

Data Source

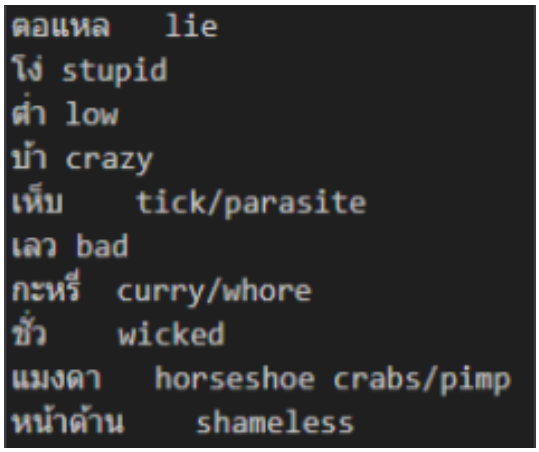
The main source of data used in this project comes from the ThaiToxicityTweetCorpus, which is an academic data set for research located on GitHub[2]. The data set was developed and is maintained by tmu-nlp(Natural Language Processing) group and is supported by academic institutions, making it an academic and professionally oriented data source for analysis of toxic language. From the data set provided, toxic_keywords.txt was selected. The data set was decided to be reliable as it was created by researchers for the purpose of identifying the toxicity in the Thai language, specifically in natural language processing.

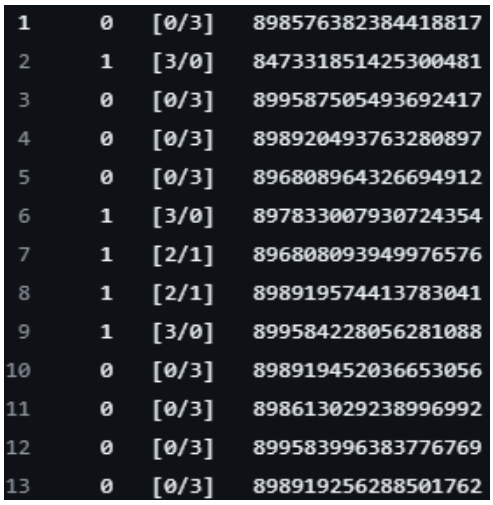
This source of data is chosen for its quality and specific purpose in this project, as the Thai language has issues that may be difficult for a model built using data for English. For instance, slang in the Thai language is varied and uses different meanings of words; Thai slang is also difficult to detect, as sarcasm often includes similar wordings. The lack of spaces between Thai words is a particular problem for a model trained on the English language, which is not found in the Thai language; however, the data source used overcame this problem. The sources are derived from online sites that reflect prevalent cyberbullying across networks like Twitter/X and Facebook, and will serve as a good source of data for identifying popular cyber bullying text. At the time of this research project, the data set had 44 Thai words defined as toxic/harassing language and used for model construction.

The sources are easy to access and available to be downloaded directly to the respective system where data will be stored, as the source data was freely available and distributed under an academic usage license from the mentioned GitHub repository. This data set is in TSV format, which represents a rigidly defined and predictable format with a standard two-column pattern of Thai terms followed by their English translation. Due to its well-defined structure, the operating system can effectively manage the input and output streams to the relevant locations by directly controlling the process with the usage of I/O management and the memory allocation process of the operating system.

Data Format

This project utilizes two files from the ThaiToxicityTweetCorpus repository.





1	0	[0/3]	898576382384418817
2	1	[3/0]	847331851425300481
3	0	[0/3]	899587505493692417
4	0	[0/3]	898920493763280897
5	0	[0/3]	896808964326694912
6	1	[3/0]	897833007930724354
7	1	[2/1]	896808093949976576
8	1	[2/1]	898919574413783041
9	1	[3/0]	899584228056281088
10	0	[0/3]	898919452036653056
11	0	[0/3]	898613029238996992
12	0	[0/3]	899583996383776769
13	0	[0/3]	898919256288501762

4.1 Format Classification

The datasets that are used in our project are both structured data. The annotated_corpus.txt and toxic_keywords.txt files are text files that are organized into distinct and predictable columns. The toxic_keywords.txt file is [Thai word] [tab] [Translation], and the annotated_corpus.txt file is [toxic/not toxic][tab][votes] [tab] [tweet_id].

4.2 Format Definition

Structured Data follows a predefined, rigid format with a clear organization. It is typically stored in tabular form (rows and columns), where each field has a defined data type.

4.3 Format Examples

The ThaiToxicityTweetCorpus is divided into two main parts.
The annotated_corpus.txt includes the toxic or non-toxic flag(0 = non-toxic, 1 = toxic), then the ratio of the votes if that message is toxic or non-toxic
(3/0 means all 3 annotators' votes that this tweet is toxic), and then the tweet ID of the message, and the pattern of the data will be [toxic/not toxic][tab][votes] [tab] [tweet_id].
The other part of ThaiToxicityTweetCorpus is the ttoxic_keywords.txt, which is structured so that each row contains Thai words as the feature for harassment with their English definitions. Examples include "ควาย" which means "buffalo/stupid" or "เหี้ย" which is a Thai vulgar word meaning "varanus salvator/damn", the pattern of the data will be [Thai toxic word][meanings].

4.4 Format Characteristics

Both of the files are in a TSV format, which is similar to a CSV file but uses a Tab separator instead of a comma. Each line of all the files has a strict pattern that is stated in the format examples.

Amount of Data

The dataset used in this project is the ThaiToxicityTweetCorpus. It includes an annotated_corpus.txt that contains Thai tweets collected from X, where each entry includes a label and an annotation ratio from three annotators.

The dataset also provides a toxic_keywords.txt file that contains 44 predefined toxic keywords used to collect the tweets through the X search API, helping capture harmful language from real-world contexts.

Since the dataset is already annotated, no additional labeling is required. The project focuses on categorizing text into toxic or non-toxic categories. The data is stored in text format, making it easy to use for text-matching algorithms.

During execution, the system processes text in real time using multi-threading, allowing evaluation of CPU scheduling and system performance under workload.

References

[1] H. Iqbal, M. Sabir, A. Razzaq, and J. Munir, "Detecting online harassment based on social media text by using ensemble learning," *Kashf J. Multidiscip. Res.*, vol. 2, no. 6, pp. 48–64, Jun. 2025. doi: 10.71146/kjmr485.

[2] tmu-nlp, "ThaiToxicityTweetCorpus," GitHub, 2018. [Online]. Available: https://github.com/tmu-nlp/ThaiToxicityTweetCorpus